# Zinc

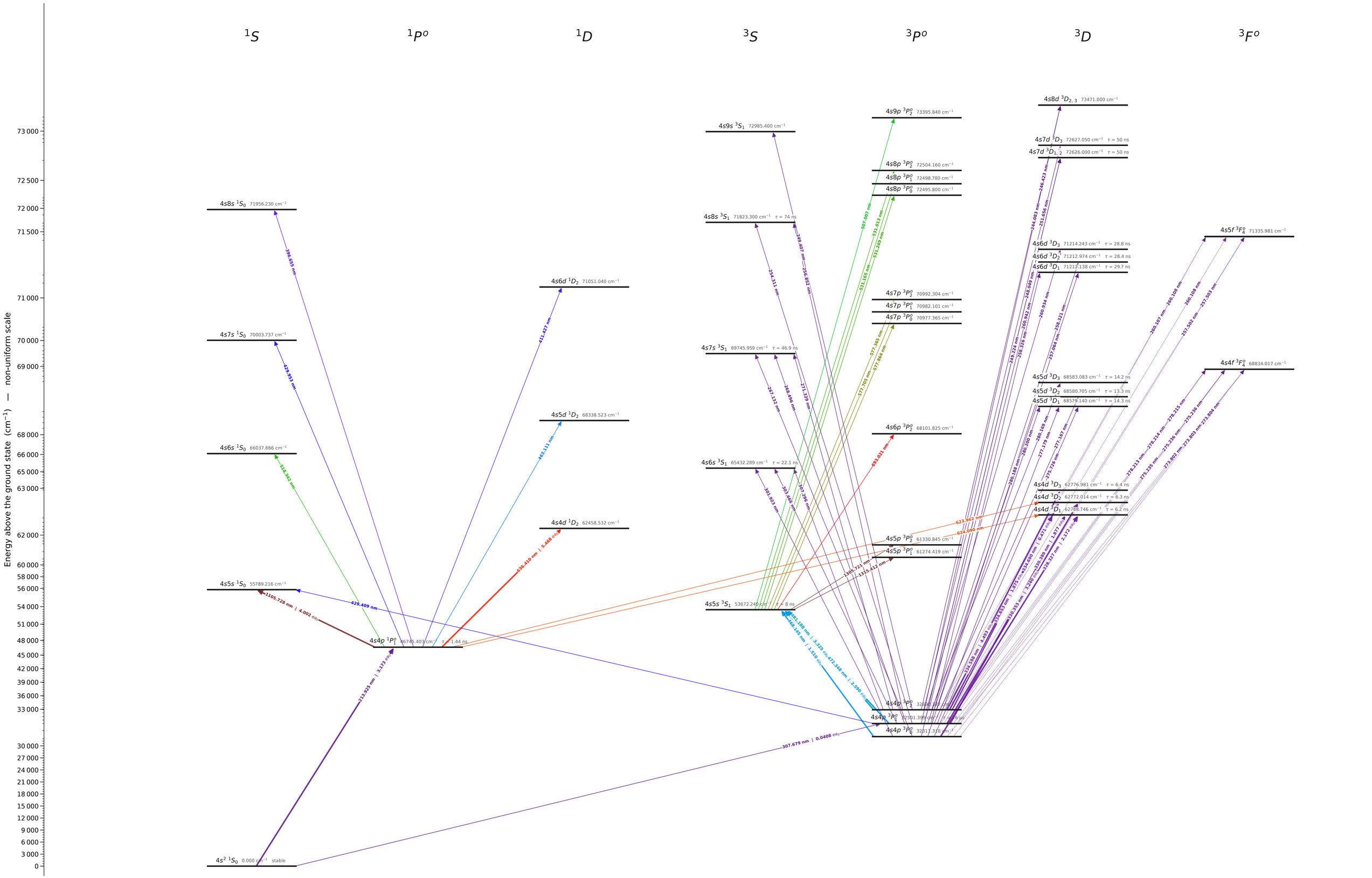
Zn (Zn I). Strongest classified transitions below 2 μm (NIST): 47 levels, 70 lines.

$Zn^+$ ionisation limit 75769.31 $cm^{-1}$ = 9.39420 eV (131.980 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in $ea_0$
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient $A$. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n$ = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000




## Key transitions

| Transition | λ vac (nm) | λ air (nm) | ν (THz) | $\|\langle J\|er\|J'\rangle\|$ ($ea_0$) | A ($s^{-1}$) | Γ/2π up. (MHz) | branch (%) | Use |
|---|---|---|---|---|---|---|---|---|
| $4s^2\ ^1S_0$ – $4s4p\ ^1P^o_1$ | 213.9248 | 213.8574 | 1401.39166(6) | 3.173 | 6.944e+08 | 110.5 | 100 | 213.9 nm resonance line: MOT / slowing / imaging |
| $4s^2\ ^1S_0$ – $4s4p\ ^3P^o_1$ | 307.6791 | 307.5897 | 974.3674 | 0.041 | 3.850e+04 | 0.006121 | 100 | 307.6 nm intercombination line: narrow-line cooling |
| $4s4p\ ^3P^o_2$ – $4s5s\ ^3S_1$ | 481.1877 | 481.0532 | 623.0261 | 3.325 | 6.700e+07 | 19.89 | 53.6 | 481.2 nm line from the metastable 4s4p 3P2 level: colline... |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## $^{67}$Zn hyperfine structure (I = 5/2)

| Level | A (MHz) | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|---|---|---|---|
| $4s4p\ ^3P^o_2$ | 608.922(3) | -18.995(10) | F=3/2: -2144.52 F=5/2: -593.73 F=7/2: +1517.56 |
| $4s4p\ ^3P^o_2$ | 531.987(5) | 35.806(5) | F=1/2: -3698.84 F=3/2: -2916.98 F=5/2: -1604.91 F=7/2: +250.78 F=9/2: +2668.89 |
| $4s4p\ ^1P^o_1$ | 20.0(20) | 10(5) | F=3/2: -63.00 F=5/2: -28.00 F=7/2: +52.50 |
| $4s5s\ ^3S_1$ | 1266.5(18) | – | F=3/2: -4432.75 F=5/2: -1266.50 F=7/2: +3166.25 |

Measured A, B constants; sources in the data file.

## Isotope shifts

| Transition | λ vac (nm) | Isotopes | Shift (MHz) |
|---|---|---|---|
| $4s^2\ ^1S_0$ – $4s4p\ ^1P^o_1$ | 213.925 | 66-64 | 525(3) |
| $4s^2\ ^1S_0$ – $4s4p\ ^1P^o_1$ | 213.925 | 67-64 | 835(4) |
| $4s^2\ ^1S_0$ – $4s4p\ ^1P^o_1$ | 213.925 | 68-64 | 1039.8(17) |
| $4s^2\ ^1S_0$ – $4s4p\ ^1P^o_1$ | 213.925 | 70-64 | 1495(4) |
| $4s^2\ ^1S_0$ – $4s4p\ ^3P^o_1$ | 307.679 | 66-64 | 689.411(3) |
| $4s^2\ ^1S_0$ – $4s4p\ ^3P^o_1$ | 307.679 | 67-64 | 1085.933(7) |
| $4s^2\ ^1S_0$ – $4s4p\ ^3P^o_1$ | 307.679 | 68-64 | 1362.390(3) |
| $4s^2\ ^1S_0$ – $4s4p\ ^3P^o_1$ | 307.679 | 70-64 | 1958.358(7) |
| $4s4p\ ^3P^o_2$ – $4s5s\ ^3S_1$ | 481.188 | 64-68 | -141.2(12) |
| $4s4p\ ^3P^o_2$ – $4s5s\ ^3S_1$ | 481.188 | 66-68 | -63.6(15) |
| $4s4p\ ^3P^o_2$ – $4s5s\ ^3S_1$ | 481.188 | 67-68 | -41.4(21) |
| $4s4p\ ^3P^o_2$ – $4s5s\ ^3S_1$ | 481.188 | 70-68 | 69.5(9) |

Shift = ν(first isotope) − ν(second isotope).